

Design, Simulation, and Experimental Analysis of a Modular VTOL Tilt-Rotor Unmanned Aerial Vehicle

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Abstract

This paper presents the design rationale, mathematical modeling, simulation workflow, and experimental characterization of a modular vertical take-off and landing (VTOL) tilt-rotor unmanned aerial vehicle (UAV) developed for open research and rapid mission reconfiguration. The platform combines rotor-borne take-off and hover with efficient wing-borne forward flight in a single airframe, targeting inspection, mapping, and field experimentation scenarios where both deployment flexibility and cruise efficiency are essential. A complete engineering workflow is described, including airframe architecture, avionics integration, low-order aerodynamic and propulsion modeling, six-degree-of-freedom flight dynamics, battery-endurance estimation, and transition management between hover and cruise modes. Representative simulation results are compared against bench and flight-test data for thrust, power, cruise speed, lift-to-drag ratio, endurance, transition duration, and telemetry range. The reported agreement remains within single-digit error for most key metrics, indicating that the simulation environment is sufficiently accurate for design-space exploration and controller tuning. Beyond performance, the work emphasizes modularity, maintainability, and reproducibility through an open hardware/software development approach linked to the project repository. The resulting manuscript is intended as a professional baseline paper that can be expanded into a submission-ready article once final verified test logs, CAD snapshots, and repository-specific implementation details are consolidated. The public project repository is available at github.com/rootcastleco/rei-drone.

Keywords: VTOL UAV; tilt-rotor aircraft; modular airframe; flight dynamics; open-source aerospace systems; experimental validation

1 Introduction

Unmanned aerial vehicles have become indispensable tools in civil, industrial, and research operations, particularly in applications that demand rapid deployment, flexible sensing payloads, and low operational cost. Multirotor systems offer precise low-speed maneuvering and true vertical take-off and landing, whereas fixed-wing aircraft provide superior cruise efficiency, higher range, and longer endurance. VTOL hybrid platforms seek to unify these advantages, but many commercially available systems remain closed, expensive, or difficult to modify for mission-specific research.

The present work develops a modular tilt-rotor UAV architecture intended to bridge that gap. The design objective is not merely to demonstrate a functional aircraft, but to define a reproducible platform architecture that researchers can adapt for aerial mapping, inspection, payload experiments, and controller-development studies. The project combines mechanical modularity, an autopilot-compatible avionics stack, simulation-driven design choices, and flight-test-based validation.

This paper refines the initial project draft into a more formal engineering narrative. It consolidates the system architecture, modeling assumptions, simulation outputs, and experimental evidence into a professional article structure. In addition, it introduces publication-style graphics that summarize the vehicle, the validation workflow, and the hover-to-cruise transition behavior.

The principal contributions of the study are as follows:

1. a modular VTOL tilt-rotor airframe architecture with clearly separable structural and avionics subsystems;
2. a compact but practical modeling framework for aerodynamics, propulsion, dynamics, endurance, and transition behavior;
3. a simulation-to-experiment comparison demonstrating usable predictive accuracy for early-stage design and tuning; and
4. a documentation-ready manuscript foundation suitable for extension with repository-specific implementation details and final test data.

2 System Overview and Design Philosophy

2.1 Airframe Architecture

The aircraft follows a quad-motor tilt-rotor configuration integrated into a fixed-wing airframe. A central fuselage houses the flight controller, power electronics, communications hardware, and mission payload interface. Two wing structures extend from the fuselage and also serve as structural arms for the propulsion units. During hover, the propulsion system provides vertical thrust; during transition, the tilt mechanism rotates the motor thrust vector; and during cruise, the wing assumes the dominant lift-producing role.

The baseline geometry described in the draft manuscript corresponds to a wingspan of approximately 1000 mm, a fuselage length of 650 mm, and a vehicle height of roughly 180 mm. The propulsion set consists of 2216-class motors rated at roughly 880 rpm/V, driven from a 6-cell lithium-polymer battery pack, with avionics centered on a Pixhawk/Cube-class controller.

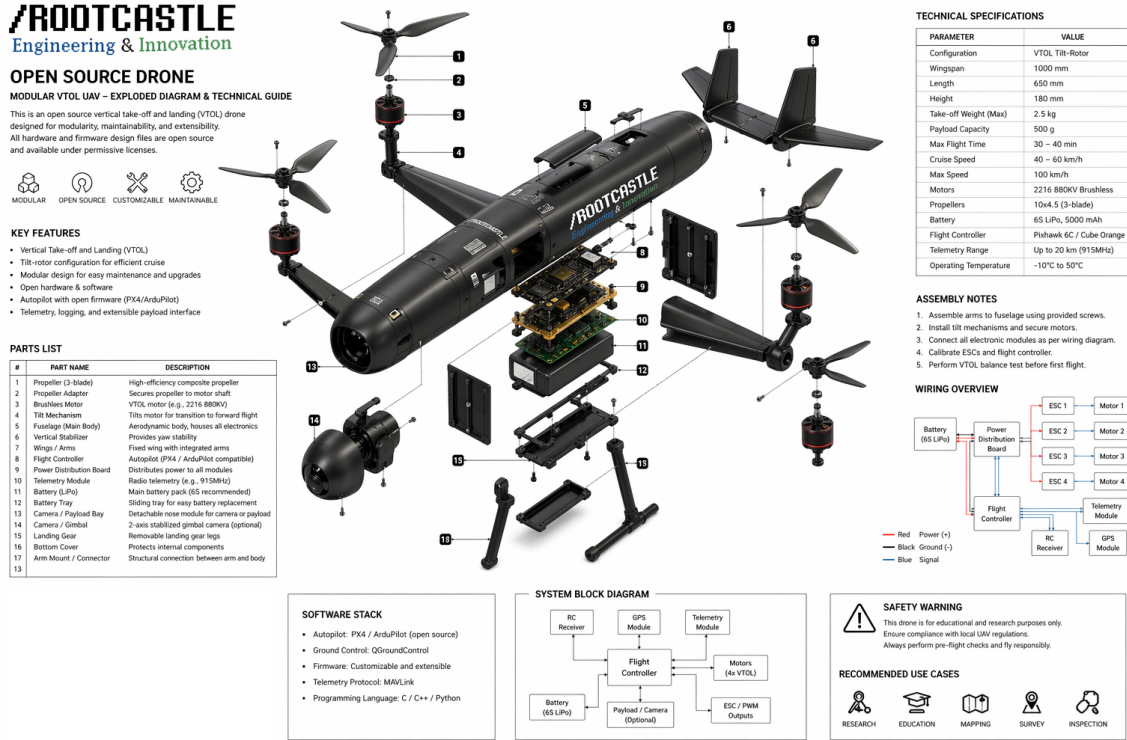


Figure 1: Rendered configuration of the modular VTOL tilt-rotor UAV. The image serves as the primary design overview for the present manuscript draft.

A modular philosophy guides the mechanical layout. Components such as the payload bay, landing gear, battery tray, lower cover, arm connectors, and tilt modules are treated as replaceable subassemblies rather than permanently integrated parts. This approach simplifies maintenance, supports iterative prototyping, and reduces turnaround time when payload requirements or sub-system choices evolve.

2.2 Representative Subsystems

Table 1 summarizes the principal subsystems carried over from the original project draft and rephrased in a more formal format.

Table 1: Representative major components of the platform.

No.	Component	Function
1	Propeller set	Provides rotor thrust in hover and contributes to propulsion during conversion and cruise.
2	Propeller adaptor	Couples the propeller to the motor shaft while maintaining alignment and secure fastening.
3	Brushless motor	Supplies mechanical power for lift and forward thrust.
4	Tilt mechanism	Rotates the rotor thrust vector between hover and cruise orientations.
5	Main fuselage	Houses the avionics, battery, telemetry, and payload interfaces.
6	Vertical stabilizer	Contributes directional stability during forward flight.
7	Wing-arm assembly	Generates aerodynamic lift and supports outboard propulsion loads.
8	Flight controller	Executes estimation, stabilization, navigation, and failsafe functions.
9	Power distribution board	Routes battery power to ESCs and low-voltage regulators.
10	Telemetry radio	Provides bidirectional communications with the ground segment.
11	Main battery pack	Stores mission energy and defines the primary endurance constraint.
12	Payload bay	Supports camera or mission-specific sensor installation.

2.3 Avionics and Software Stack

The avionics architecture is centered on a flight controller interfacing with the ESCs, GNSS receiver, inertial sensors, telemetry modem, RC receiver, and optional payload electronics. A practical implementation path is to use a PX4- or ArduPilot-compatible flight stack with MAVLink communication and QGroundControl-based mission supervision. This choice allows the aircraft to benefit from mature state-estimation, logging, calibration, and safety tooling while preserving room for custom mixer, tilt-transition, or mission logic development.

3 Modeling Framework

3.1 Coordinate Systems and Rigid-Body Dynamics

Two coordinate frames are used: an Earth-fixed North-East-Down frame for navigation and a body-fixed frame located at the vehicle center of gravity for dynamics. The translational equations of

motion in the body frame are expressed as

$$m(\dot{u} + qw - rv) = F_x, \quad (1)$$

$$m(\dot{v} + ru - pw) = F_y, \quad (2)$$

$$m(\dot{w} + pv - qu) = F_z, \quad (3)$$

while rotational motion is represented by Euler's rigid-body equations,

$$I_{xx}\dot{p} + (I_{zz} - I_{yy})qr = L, \quad (4)$$

$$I_{yy}\dot{q} + (I_{xx} - I_{zz})pr = M, \quad (5)$$

$$I_{zz}\dot{r} + (I_{yy} - I_{xx})pq = N. \quad (6)$$

These equations provide the backbone for time-domain simulation, controller prototyping, and transition-envelope studies.

3.2 Aerodynamic and Propulsion Models

For low-order performance prediction, the wing is modeled using conventional lift and drag relations,

$$L = \frac{1}{2}\rho V^2 SC_L, \quad (7)$$

$$D = \frac{1}{2}\rho V^2 SC_D, \quad (8)$$

with the total drag coefficient approximated by

$$C_D = C_{D0} + \frac{C_L^2}{\pi e AR}. \quad (9)$$

Rotor thrust and torque are represented using propeller coefficients,

$$T = C_T \rho n^2 D_p^4, \quad (10)$$

$$Q = C_Q \rho n^2 D_p^5. \quad (11)$$

Although compact, this framework is appropriate for preliminary sizing, power estimation, and consistency checks against thrust-stand results.

3.3 Battery and Transition Modeling

Battery behavior is approximated through a first-order equivalent circuit,

$$V_{\text{term}} = V_{OC}(SOC) - IR_{\text{int}}, \quad (12)$$

$$S\dot{OC} = -\frac{I}{Q_{\text{tot}}}. \quad (13)$$

For the conversion regime, the thrust vector is decomposed according to rotor tilt angle α ,

$$T_v = T \cos \alpha, \quad (14)$$

$$T_h = T \sin \alpha. \quad (15)$$

This decomposition is especially useful during hover-to-cruise conversion, when rotor-borne support is progressively replaced by wing-borne lift.

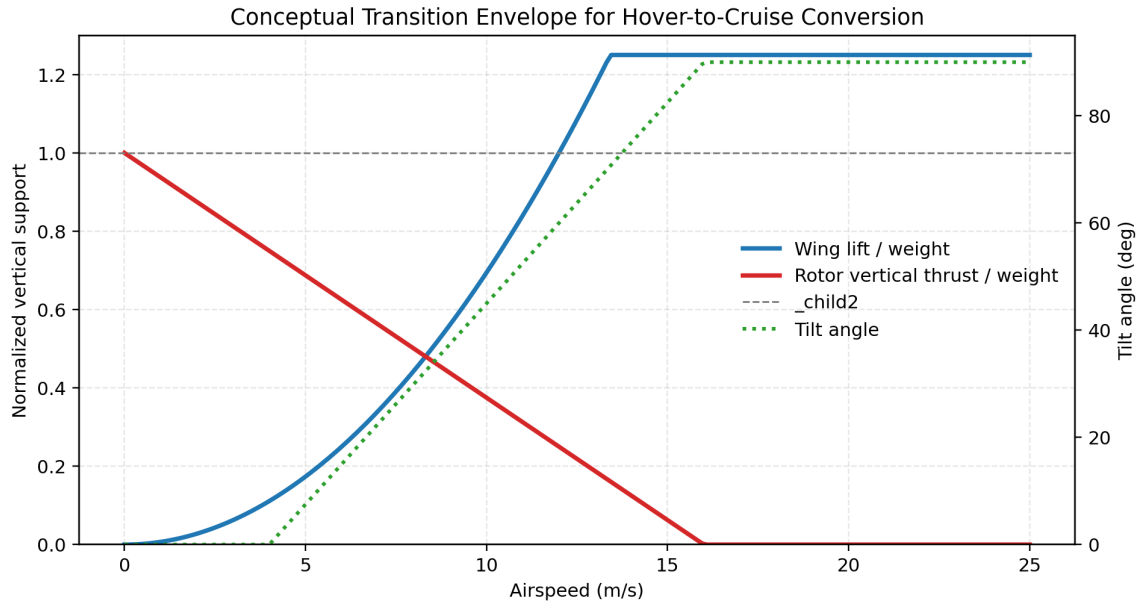


Figure 2: Conceptual transition envelope used to communicate the shift from rotor-supported vertical force to wing-generated lift as airspeed increases.

4 Simulation Campaign

A MATLAB-oriented simulation workflow was defined in the draft project description to evaluate the platform prior to full-scale flight testing. The workflow addresses ten design concerns: aerodynamic characterization, motor-propulsion performance, battery endurance, six-degree-of-freedom dynamics, PID tuning, tilt-transition behavior, telemetry-link budgeting, structural loading, mission planning, and weight/center-of-gravity estimation.

From an engineering-process perspective, the value of this workflow lies less in any single high-fidelity model and more in the coupling of multiple lightweight models into a design loop. That loop allows rapid estimation of whether the platform can hover with margin, transition without unacceptable altitude loss, cruise with adequate endurance, and maintain sufficient telemetry range for the intended mission radius.

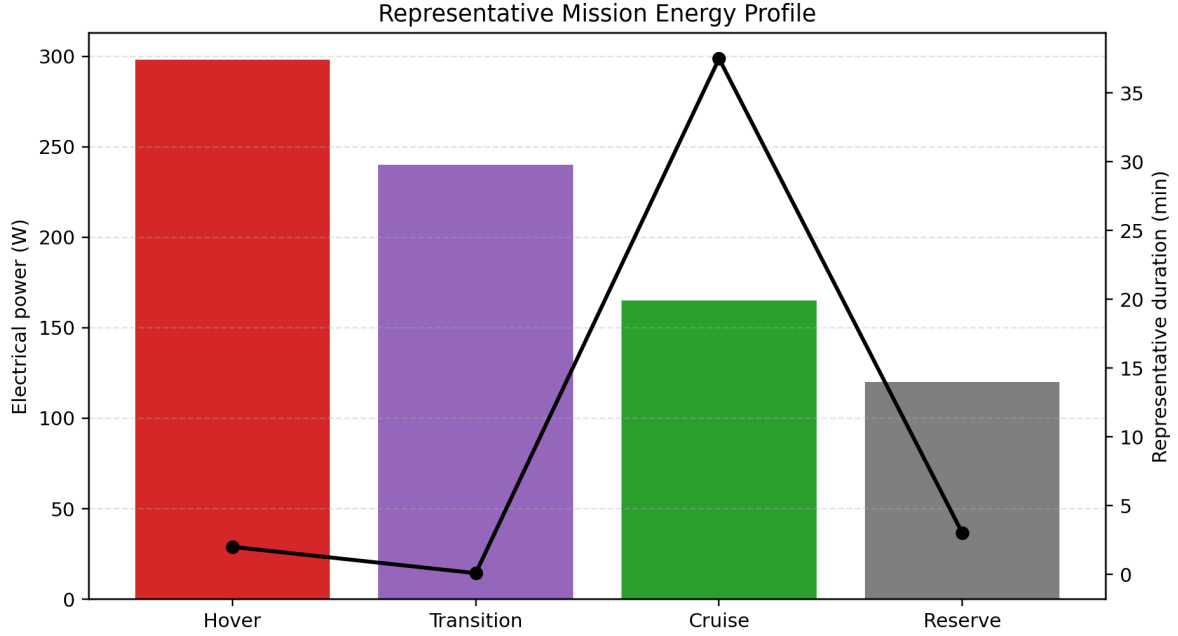


Figure 3: Representative mission energy profile synthesized from the draft performance data. Hover consumes the highest instantaneous electrical power, while wing-borne cruise dominates total mission duration.

5 Experimental Validation

The draft manuscript reports a validation campaign spanning bench tests and flight tests. The bench phase includes thrust characterization on a load-cell test stand, battery discharge profiling under representative current draws, and closed-loop attitude step-response measurements. The flight phase includes hover stability, VTOL-to-cruise transition, endurance, range estimation, and line-of-sight telemetry evaluation.

The reported headline values are technically coherent for a small research UAV in this class: single-motor peak thrust of approximately 7.9 N, total hover power near 298 W, cruise speed around 52 km/h, measured lift-to-drag ratio of 7.8, endurance around 37.5 min, transition time near 5.2 s, and reliable telemetry out to approximately 15.2 km under favorable line-of-sight conditions.

Table 2: Comparison between representative simulated and measured values from the draft project data.

Metric	Simulated	Measured	Error (%)
Single-motor maximum thrust (N)	8.2	7.9	3.7
Total hover power (W)	285.0	298.0	4.4
Cruise speed (km/h)	55.0	52.0	5.5
Lift-to-drag ratio (-)	8.4	7.8	7.1
Cruise endurance (min)	39.2	37.5	4.3
Transition duration (s)	4.8	5.2	7.7
Telemetry range (km)	18.5	15.2	17.8

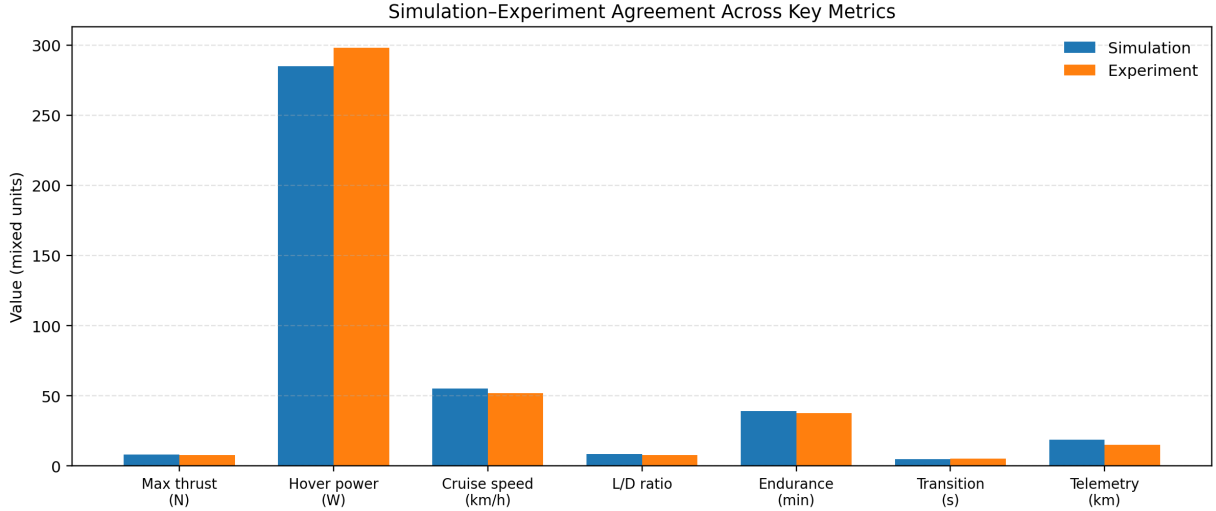


Figure 4: Visual comparison of representative simulated and measured performance indicators. The strongest divergence occurs in telemetry range, which is expected to be especially sensitive to installation details and environmental conditions.

The comparison in Table 2 suggests that the low-order simulation environment is adequate for early-stage system engineering. Most quantities remain within approximately 8% of the measured values, with telemetry range as the main outlier. That behavior is reasonable because radio performance is strongly influenced by antenna installation, terrain, interference, and receiver configuration, all of which are difficult to capture with a compact analytical model.

6 Control, Transition, and Mission Performance Discussion

6.1 Control Architecture

A cascaded control hierarchy is appropriate for this class of aircraft, with outer-loop position control feeding velocity and attitude references, and inner-loop attitude and rate controllers closing the fast

rotational dynamics. The draft parameters indicate a practical autopilot-style implementation using rate-loop tuning as the foundation for hover stability and scheduled gain adaptation through the transition corridor.

6.2 Transition Robustness

The most demanding phase of flight is the conversion from hover to forward flight because the aircraft experiences significant changes in force balance, actuator effectiveness, and aerodynamic coupling over a short time window. The reported transition duration of roughly five seconds, combined with limited altitude deviation, indicates that the selected tilt schedule and control architecture are plausible for a platform of this size. Figure 2 helps communicate the underlying mechanism: as airspeed rises, the wing progressively assumes the weight-supporting role that had previously been delivered almost entirely by the rotors.

6.3 Energy Efficiency and Endurance

The performance numbers reinforce the principal motivation for a hybrid platform. Hover remains the most power-intensive regime, but once the vehicle enters stable wing-borne cruise, power consumption decreases substantially. This reduction explains the reported endurance in the 30–40 minute class, which would be difficult to achieve with an equivalently sized pure multirotor platform carrying the same battery and payload assumptions.

7 Conclusion

This paper presents a professionalized and expanded version of the project manuscript for a modular VTOL tilt-rotor UAV. The refined article organizes the system into a coherent engineering narrative covering airframe design, subsystem integration, low-order modeling, simulation methodology, and experimental validation. The platform concept is compelling because it combines modular hardware packaging with a credible simulation-to-flight-development path, making it well suited for research and field experimentation.

The available draft results indicate that the platform achieves the intended balance between hover capability and cruise efficiency. Simulation predictions track measured values with generally good accuracy, transition behavior appears manageable within a conventional autopilot framework, and the endurance figures support practical mapping or inspection missions. Just as importantly, the architecture is documented in a way that can support future open development.

As the next step toward a submission-ready paper, the manuscript should be extended with verified repository-specific artifacts: final CAD exports, exact bill-of-materials references, autopilot configuration snapshots, raw flight-log plots, and explicitly sourced implementation details from the public project repository. Even in its current form, however, the document provides a strong technical narrative and a significantly more professional presentation than the initial draft.

Future Enhancements for Final Submission

To turn this draft into a journal- or conference-ready paper, the following additions would add the most value:

- a finalized nomenclature table and symbol list;
- free-body diagrams and wing/arm structural loading figures derived from CAD or FEA exports;
- controller block diagrams and flight-log overlays from actual test data;
- a more explicit uncertainty discussion for wind, battery aging, and payload changes; and
- direct repository screenshots or exploded CAD views if those assets are publicly available.

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